$$\underline{s} = \underline{p}_2 - \underline{p}_1 ; \quad l = \|\underline{s}\| = \sqrt{\underline{s}^T \underline{s}} ; \quad \underline{p} = [\underline{p}_1^T \; \underline{p}_2^T] ; \quad \underline{d} = \frac{\underline{s}}{l} ;$$

$$\underline{f} = k(l - l_0)\underline{d} = \underline{s} \cdot k \cdot \left(1 - \frac{l_0}{l}\right) ;$$

$$\underline{\underline{J}} = \frac{\partial \underline{f}}{\partial \underline{p}} = k \cdot \frac{\partial \underline{s} \cdot \left(1 - \frac{l_0}{l}\right)}{\partial \underline{p}} = k \cdot \left[\frac{\partial \underline{s}}{\partial \underline{p}} \cdot \left(1 - \frac{l_0}{l}\right) - l_0 \cdot \underline{s} \frac{\partial l^{-1}}{\partial \underline{p}} \right]$$

$$\Leftarrow \frac{\partial l^{-1}}{\partial \underline{p}} = \frac{\partial \left(\sqrt{\underline{s}^T \underline{s}}\right)^{-1}}{\partial \underline{p}} = -\frac{1}{2\sqrt{(\underline{s}^T \underline{s})^3}} \cdot \frac{\partial \underline{s}^T \underline{s}}{\partial \underline{p}} = -\frac{1}{2l^3} \cdot \frac{\partial \underline{s}^T \underline{s}}{\partial \underline{p}} =$$

$$\Rightarrow \underline{s}^T \underline{s} = (\underline{p}_2 - \underline{p}_1)^T (\underline{p}_2 - \underline{p}_1) = (p_{2x} - p_{1x})^2 + (p_{2y} - p_{1y})^2 + (p_{2z} - p_{1z})^2 ;$$

$$\Rightarrow \frac{\partial \underline{s}^T \underline{s}}{\partial \underline{p}} = 2 \begin{bmatrix} p_{2x} - p_{1x} \\ p_{2y} - p_{1y} \\ \dots \\ -p_{2x} + p_{1x} \\ \dots \end{bmatrix}^T = 2 [-\underline{s}^T \; \underline{s}^T] ;$$

$$\frac{\partial \underline{s}}{\partial \underline{p}} = \left[-\underline{\underline{I}} \quad \underline{\underline{I}} \right] \Leftarrow$$

$$\underline{\underline{J}} = k \cdot \left(1 - \frac{l_0}{l}\right) \left[-\underline{\underline{I}} \quad \underline{\underline{I}} \right] + \frac{k \cdot l_0}{l^3} \underline{s} \cdot [-\underline{s}^T \quad \underline{s}^T] =$$

$$= k \cdot \begin{bmatrix} -\frac{l_0}{l^3} \underline{s}\underline{s}^T - \left(1 - \frac{l_0}{l}\right) \underline{\underline{I}} \\ \\ \frac{l_0}{l^3} \underline{s}\underline{s}^T \quad \left(1 - \frac{l_0}{l}\right) \underline{\underline{I}} \end{bmatrix}^T$$